

RAJESWARI M

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Summary

DevOps and Machine Learning-focused engineer with hands-on experience in building scalable systems, automating workflows, and developing ML-based solutions. Proficient in Python, containerization, and backend development, with experience in deploying data-driven applications and designing reliable, secure systems.

Education

Kongu Engineering College Nov 2025 – Present
B.Tech in Information Technology CGPA: 8.37/10

Omega International School, Chennai 2023
Higher Secondary (CBSE) 90.8%

Technical Skills

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| <ul style="list-style-type: none">• Programming: Python, Java• DevOps: Docker, CI/CD Pipelines, GitHub Actions• Cloud: Cloud fundamentals• Systems: Linux, Bash | <ul style="list-style-type: none">• Machine Learning: Scikit-Learn, Neural Networks• Tools: Git, GitHub• Backend: Node.js, Express• Databases: MongoDB, PostgreSQL, Oracle DB |
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Experience

National Technical Research Organization New Delhi, India
Intern – Machine Learning & Cryptography Research Apr 2025 – Sep 2025

- **Cipher Systems Research**
 - Analyzed classical and mechanical cipher systems, studying their design, operational mechanisms, and inherent security limitations.
 - Reverse engineered HC-series cipher machines to understand internal structures and identify potential vulnerabilities.
- **ML-Based Cipher Classification**
 - Developed an end-to-end machine learning pipeline for cryptographic algorithm classification using datasets generated from binary files and curated repositories.
 - Applied feature engineering using NIST statistical test suites to extract distinguishing cryptographic patterns.
 - Evaluated multiple ML models and built an optimized classifier for accurate cipher identification.
- **Automated Reasoning System**
 - Built a graph-based reasoning system using Neo4j to model relationships between cryptographic entities and enable structured analysis.
 - Integrated a rule-based inference layer to enhance logical reasoning and support path discovery.
 - Validated the system on classical ciphers (Caesar, Playfair, Vigenère), ensuring consistent classification under controlled conditions.
- **Repository:** https://github.com/Rajeshwarimughilan/CipherDetection_ResearchWork.git
- **Technical Report:** Detailed implementation, experiments, and outputs available at [View Report](#)

Projects

TorpedoX (Project Link)

April 2025

Developed an ML-based cryptographic analysis tool using **Python, PyTorch, Scikit-learn, NumPy, and Pandas** to identify cryptographic algorithms and hash functions.

Designed and trained **FFNN** and **Random Forest** models with feature extraction pipelines to classify classical and modern ciphers with strong generalization.

Evaluated multiple ML models and optimized feature extraction pipelines to improve classification performance across cipher types.

Classical Cryptanalysis: Kappa Test & Crib Dragging (Project Link)

July 2025

Built a cryptanalysis pipeline in **Python** to detect keystream reuse using **Kappa Index of Coincidence**.

Implemented an interactive crib dragging tool with output persistence for plaintext recovery and ciphertext alignment.

Applied statistical analysis techniques to identify patterns and improve efficiency of cipher-only attack workflows.

Reference: Stallings — *Cryptography and Network Security*, Chapter 10.

Loyalo CMS Platform (Project Link)

Present

Built a CMS-driven web application using **React, Node.js, Express, SQLite (Prisma)** with dynamic content delivery via RESTful APIs.

Designed secure admin workflows with **JWT authentication, validation, and rate limiting** for scalable content management.

Designed RESTful APIs for dynamic content delivery and efficient admin-side operations.

Achievements

- **Smart India Hackathon 2024** — National Winner
 - **Joint Secretary, IT Association**
 - **Student POC, Innovation Forum**
- Nov 2025 – Present

Jan 2025

Certifications

- **GUVI DevOps Certification**
- **NPTEL: Design & Implementation of Human–Computer Interfaces**